A 2.5D-Aware Analytical Placement Framework for Non-Planar Printed Circuit Board Designs

Abstract

As board area is limited and component density rises, modern 2.5D non-planar printed circuit board (PCB) designs increasingly contain dense modules, multi-height components, and non-planar geometries that challenge conventional layout flows. Standard 2D analytical placers misclassify valid vertical nesting as planar overlap and fail to distinguish between projected conflicts and true physical collisions, thereby unnecessarily restricting component placement and potentially increasing connection length. We present, to the best of our knowledge, the first 2.5D-aware analytical placement framework for non-planar PCB designs. Our framework integrates mesh-based collision reasoning into an electrostatic density formulation by using the density grid for bin-level pruning and applying a lightweight V-HACD/GJK-based narrow-phase collision test only to spatially relevant component pairs. The resulting 2.5D collision-aware density model exempts volumetrically separable projected overlaps while preserving the optimization backbone of conventional analytical placers. We further incorporate a 2.5D-aware legalization stage to consistently handle true collisions after global placement. Evaluations on seven industry-grade PCB benchmarks show that our method reduces true physical collisions to 13% of the conventional 2D baseline, while identifying substantial valid vertical nesting opportunities. Because projected overlap, true physical collision, and connection length capture different aspects of a 2.5D placement, XHPWA is reported as a secondary placement-quality metric rather than as a standalone measure of physical feasibility. The results demonstrate that explicit 2.5D geometric reasoning can substantially improve physical feasibility and enable more compact local placement through legal vertical nesting, with additional placement runtime that remains practical on the evaluated benchmarks.

1 Introduction

Printed circuit board (PCB) designs are rapidly evolving toward compact, multi-height, and non-planar 2.5D assemblies. Modern systems integrate stacked sensor modules, daughterboards, shielding structures, and mechanical frames that occupy distinct Z-levels while sharing overlapping projected regions on the board. As these assemblies become denser and increasingly intertwined in 3D space, manual placement becomes impractical, motivating the need for automated global placement tools that can reason about spatial feasibility. In this work, we use the term "2.5D" to denote a placement setting in which component positions are optimized in the board plane while full 3D component geometry is retained for collision and vertical-clearance evaluation.

Analytical global placement is widely adopted in IC and PCB design flows because it formulates placement as a continuous optimization problem solved through gradient-based methods. However, current analytical placers for PCBs remain fundamentally planar: components are represented by 2D rectangles or polygons, density is defined by their projected area on the board, and configurations exhibiting excessive overlap are penalized. Because the formulation operates purely on 2D density, it cannot distinguish true 3D interference from vertically permissible nesting — an inherent mismatch with modern multi-height assemblies. In practice, this leads to an overly conservative use of board area, restricting the design space that the placer is allowed to explore.

Figure 1 shows a representative example of planar misclassification. In a 2D analytical placer, the projected outlines cause the red component to appear to overlap with the neighboring module, preventing it from being placed underneath. In reality, their full 3D geometry provides sufficient vertical clearance and forms a collision-free nesting arrangement for the illustrated components. With the awareness of the 3D nesting structure, the combined red connections can be shortened from 12.97 mm in Figure 1(c) to 6.51 mm in Figure 1(d), corresponding to a 6.46 mm (49.81%) reduction for the illustrated connections. Such vertical interlocking is increasingly common in modern PCB assemblies that utilize height differences among shields, connectors, and structural fixtures; however, it cannot be captured by a purely planar model.

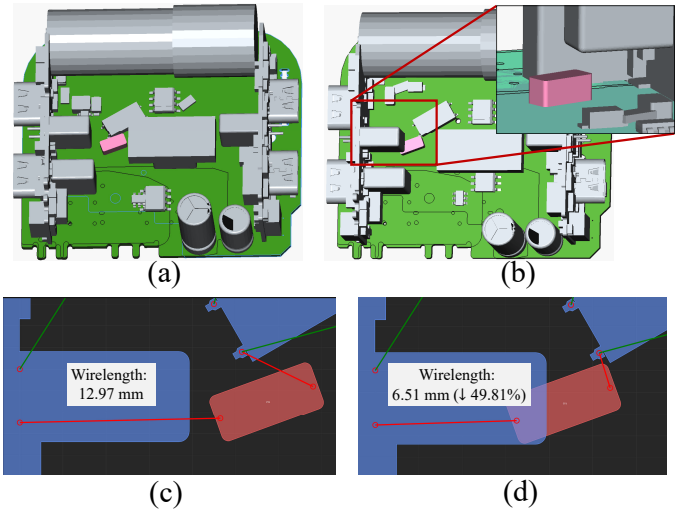


Figure 1: An example of component nesting captured from an industry-level PCB placement. (a) A 2D analytical placer cannot capture the legal 3D nesting structure with the red component underneath another component because the 2D projection shows illegal overlapping components. (b) In 3D, the two components have sufficient vertical clearance, making nesting feasible. (c) In the 2D top-down projection, this restriction forces the red component to stay farther away, resulting in longer combined red connections of 12.97 mm. (d) With 3D collision awareness, vertical nesting becomes possible, enabling tighter 2D placement and shortening the combined red connections to 6.51 mm, corresponding to a 6.46 mm (49.81%) reduction for the illustrated connections.

Beyond such a misclassification, simplified geometric modeling further introduces geometric distortion. When intricate 3D shapes are replaced by bounding boxes, convex padding, or height-inflated prisms, essential geometric detail is lost and the occupied volume becomes heavily exaggerated. As shown in Fig. 2, coarse approximations mislead the placer's interpretation of spatial feasibility by producing spurious intersections with nearby components and masking actual clearance, restricting the solution space far more than real manufacturing constraints.

At the opposite extreme, mechanical CAD and robotics systems rely on high-precision 3D collision-detection pipelines that integrate hierarchical spatial pruning with narrow-phase geometric evaluation; although accurate, these pipelines are too expensive to invoke repeatedly within an analytical placement loop, which may require thousands of density and gradient evaluations. Bridging the gap between these two regimes — oversimplified 2D models and heavyweight mechanical solvers — motivates the scalable placement framework we develop here, which incorporates lightweight but accurate 3D geometric reasoning into the density formulation, enabling 2.5D-aware and computationally efficient placement for modern 2.5D PCB designs.

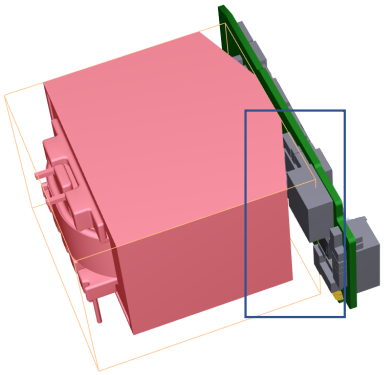


Figure 2: A snippet of local placement extracted from an industry-grade PCB layout. A bounding-box approximation (yellow box) considerably overestimates the volume of the red component, leading to spurious intersections with neighboring elements (highlighted in blue) and masking the clearance essential for valid 3D placement.

1.1 Previous Works

Research on PCB placement spans several methodological directions, including genetic algorithms, pattern-based reference placement, and analytical optimization. Genetic algorithms encode component coordinates and orientations as chromosomes and optimize heuristic objectives such as thermal performance or layout compactness through randomized search processes [1, 2]. Reference-placement methods match new designs to existing layout patterns through subgraph matching or similarity retrieval [3]. Analytical optimization methods form the core of recent academic placers. Early force-directed formulations [4] introduced continuous optimization concepts that later evolved into modern gradient-based analytical approaches for PCB layout. Subsequent works proposed net-separation regularization [5], clearance-aware force-directed spreading [6], and clearance-constrained optimization for heterogeneous components [7]. More recent developments include GPU-accelerated bilevel optimization inspired by DREAMPlace (Cypress) [8] and graph-neural-network–based spatial reasoning for component positioning [9]. Prior research in other "3D placement" domains similarly does not address the PCB setting. 3D VLSI floorplanning models modules as cuboids and optimizes their packed volume using structures such as T-trees [10], 3D-subTCG [11], weighted directed graphs [12], or 3D O-Trees [13], often within simulated-annealing frameworks [14]. 3D packing and virtual-scene object placement aim to arrange objects efficiently in space through geometric heuristics or constrained search [15, 16].

Beyond PCB-specific methods, several research directions in 3D IC placement explore vertical integration using analytical or partition-based formulations. Pseudo-3D approaches emulate 3D behavior using 2D prototypes: Compact-2D [17] derives die partitioning from a planar placement, while Snap-3D [18] transforms a 3D design into a single 2D placement structure via row remapping and tailored constraints. True-3D analytical placers such as ePlace-3D [19] extend electrostatics-based optimization into three dimensions for TSV-based 3D ICs, and more recent frameworks for heterogeneous dies [20, 21] further integrate tier assignment and inter-die interconnect costs; these placers optimize full 3D block positions, die partitioning, and inter-tier interconnect insertion within the constraints of IC manufacturing.

Although all of these domains involve three-dimensional geometry, the nature of their problems is fundamentally different from PCB design: 3D floorplanning packs cuboids to minimize overall volume, 3D IC placement distributes cells across device tiers, and 3D packing lets objects occupy arbitrary positions without electrical or manufacturing constraints. By contrast, PCB components are attached to either side of a fixed board while their geometries extend in three dimensions, so they may overlap in 2D projection yet remain mechanically compatible due to vertical clearance, or appear legal in 2D while their true 3D envelopes collide—issues not expressible in 2D analytical models nor handled by existing 3D placement formulations, which target entirely different objectives and constraints.

Table 1: Qualitative comparison with representative prior work. The "Proj./True Collision" column indicates whether a method distinguishes projected (2D) overlap from true (3D) physical collision, rather than treating every projected overlap as a physical conflict.

Method	Mesh Geom.	Proj./True Collision	Analytical Density	PCB Domain
Genetic/heuristic [1, 2]	✗	✗	✗	✓
Clearance-aware 2D [7]	✗	✗	✓	✓
3D floorplanning [10, 13]	✗	✗	✗	✗
ePlace-3D [19]	✗	✗	✓	✗
Compact-2D / Snap-3D [17, 18]	✗	✗	✗	✗
Ours	✓	✓	✓	✓

Existing analytical PCB placers therefore remain fundamentally planar: components are represented using 2D polygons or bounding rectangles, and overlap is evaluated only in projection space, preventing any reasoning about vertical separation or true volumetric feasibility. Even the most shape-aware 2D framework, the clearance-constrained global placer [7], operates purely in 2D and thus cannot exploit benign vertical nesting—the gap our framework addresses (Section 1).

Table 1 summarizes how the proposed framework relates to representative prior work along four qualitative dimensions: whether the method reasons about true 3D/mesh-level geometry (as opposed to idealized 2D polygons or cuboids), whether it distinguishes benign projected overlap from true physical collision, whether the collision reasoning is embedded directly in an analytical (gradient-based) density formulation, and whether the target domain is PCB placement specifically. To the best of our knowledge, no prior method satisfies all four properties simultaneously, which motivates the design of our framework.

1.2 Our Contributions

This paper presents a new 2.5D-aware analytical placement framework that augments a standard Poisson-based density model with mesh-level 3D collision reasoning. The main contributions are as follows:

- To the best of our knowledge, this is the first analytical PCB global placement framework that integrates mesh-based 3D collision reasoning into density optimization. Component geometry is obtained from STEP/OBJ models and converted to surface meshes. The 2D optimization operates on their board-plane placement, while the original 3D mesh geometry is retained for collision evaluation.
- We develop a bin-pruned 2.5D collision pipeline that reuses the density grid as a spatial index. Candidate component pairs are generated only when their projected footprints share density bins, avoiding unnecessary global pairwise collision checks.
- We employ a lightweight narrow-phase collision evaluation based on V-HACD convex decomposition and Gilbert–Johnson–Keerthi (GJK) collision detection. This design provides practical 3D collision reasoning inside the analytical placement loop while avoiding repeated expensive CAD-level geometric queries.
- We introduce a 2.5D collision-aware density model that exempts volumetrically separable projected overlaps. This allows the placer to exploit legal vertical nesting while continuing to penalize true 3D collisions.
- We extend the legalization stage with 2.5D-aware collision checking so that the post-placement refinement remains consistent with the 2.5D-aware global placement objective. Experimental results show

that our method reduces true 3D collisions to 13% of the 2D baseline, while achieving 0.98× the XHPWA of the conventional 2D baseline.

The remainder of this paper is organized as follows. Section 2 reviews the background and preliminaries. Section 3 describes the proposed analytical placement method for 2.5D non-planar PCB designs, including preprocessing, initial placement, global placement, and 2.5D-aware legalization. Section 4 reports the experimental results, comparing 2D and 2.5D collision-aware placement and presenting an ablation study on 2.5D-aware legalization. Section 5 concludes the paper.

2 Preliminaries

This section reviews analytical global placement and the electrostatic density model underlying our framework, introduces the X-architecture wire-area metric, and formulates the 2.5D-aware PCB placement problem.

2.1 Analytical Global Placement

We consider a PCB populated with a set of components $C = C_m \cup C_f$, where C_m and C_f denote movable and fixed components. Each movable component $i \in C_m$ is associated with a 2D placement variable $\mathbf{x}_i = (x_i, y_i)$, an in-plane orientation θ_i, and a set of pads P that are connected through nets N. We collect all movable coordinates into $\mathbf{x} = \{\mathbf{x}_i \mid i \in C_m\}$, and discretize the board outline into a density grid as described next. Analytical global placement formulates these component locations as continuous variables and optimizes a smooth objective function that approximates routing cost and geometric constraints.

The placement engine minimizes an objective of the form

$$F(\mathbf{x}) = W(\mathbf{x}) + \lambda_1 S(\mathbf{x}) + \lambda_2 D(\mathbf{x}), \tag{1}$$

where $W(\mathbf{x})$ models wire cost (e.g., wirelength or wire-area), $S(\mathbf{x})$ encodes spacing-related penalties (such as pad-to-pad clearance), and $D(\mathbf{x})$ measures density overflow in the placement region. In this work, $W(\mathbf{x})$ specifically denotes the X-architecture half-perimeter wire-area (XHPWA), i.e., the net-wise X-architecture half-perimeter wirelength multiplied by each net's routing width. The weights λ_1 and λ_2 balance routing quality against geometric feasibility. In practice, these weights may be adjusted during optimization to initially emphasize global spreading and later focus on fine-grained clearance satisfaction.

Modern analytical placers derive smooth gradients ∇W, ∇S, and ∇D and employ iterative gradient-based optimizers to update $\mathbf{x}$. Representative works such as ePlace and ePlace-MS [22] solve the density subproblem through an electrostatic analogy and use Nesterov-accelerated gradient descent to achieve fast convergence on large-scale IC benchmarks. Our method follows this paradigm but extends the density formulation to account for 3D geometry and collision awareness in PCB designs with multi-height components.

2.2 Electrostatic Density Model

The electrostatic density model discretizes the placement region into a regular grid of bins B, where each bin $b \in B$ has its area A_b. Let $A_i(b)$ denote the area of component i that falls into bin b after projection onto the xy-plane. The raw bin density is

$$\rho_b = \frac{1}{A_b} \sum_{i \in C} A_i(b), \tag{2}$$

and any excess over a target density (typically 1) represents overflow. The total overflow is

$$\sum_{b \in B} \max\big(0,\ \rho_b - 1\big) A_b, \tag{3}$$

and the density term $D(\mathbf{x})$ is chosen to approximate this overflow in a differentiable manner.

In electrostatic formulations, ρ_b is interpreted as a charge distribution over the placement region. The electric potential ϕ is obtained by solving a discrete Poisson equation of the form

$$\nabla^2 \phi = \rho - 1, \tag{4}$$

subject to Neumann or mixed boundary conditions. Efficient solvers such as ePlace-MS [22] evaluate ϕ and its gradients via fast Fourier transforms (FFTs), yielding the electric field $\mathbf{E} = -\nabla\phi$ with components (ξ_x, ξ_y) that acts as a smooth spreading force.

The field value at bin b is denoted by $\mathbf{E}_b$. The density objective can be written as

$$D(\mathbf{x}) = \tfrac{1}{2} \sum_{b \in B} (\rho_b - 1)\, \phi_b A_b, \tag{5}$$

and the gradient of D with respect to each component's position is obtained by accumulating electric-field contributions over the bins touched by that component. This construction naturally distributes repulsive forces according to the amount of excess density, so bins that are close to being full exert only mild pressure, while heavily overfilled regions strongly push components away.

Our implementation follows this electrostatic framework, and the two-sided density engine treats the top and bottom sides of the PCB as independent placement regions that share the same bin grid but maintain separate density maps. Fixed components, pads, board keepout regions, and filler cells are rasterized into B to define the target density distribution, while movable components contribute to ρ_b during each iteration. The proposed 2.5D-aware extension (Sec. 3) modifies only how $A_i(b)$ is computed and how benign overlaps are exempted, while preserving the underlying Poisson-FFT pipeline.

2.3 X-Architecture Half-Perimeter Wire-Area

Traditional wirelength models in analytical placement rely on Manhattan half-perimeter wirelength (HPWL), which assumes orthogonal routing. However, PCB routing often uses 135° segments, and X-architecture wire models provide a more faithful first-order estimate of routing span. Chen *et al.* [23] introduced an effective X-architecture placement framework in which each net is modeled by an X-shaped bounding structure whose extent captures both horizontal/vertical and diagonal routing possibilities.

We therefore use the X-architecture half-perimeter wire-area (XHPWA) as the primary wire-cost term. For a net $e \in N$, let $\mathrm{XHPWL}(e)$ denote its X-architecture half-perimeter wirelength (i.e., the length of the X-shaped bounding structure that encloses all its pins), and let w_e be its routing width. The corresponding XHPWA is then

$$A_X(e) = w_e \cdot \mathrm{XHPWL}(e).$$

The total wire-area objective $W(\mathbf{x})$ is computed by summing these XHPWA contributions over all nets and modulating them with clearance-dependent factors, as described below.

2.4 Problem Formulation

For PCB placement, the goal is to assign valid positions and orientations to irregular and multi-height components while ensuring that spacing rules are satisfied among components, pads, and mechanical features. Modern PCB designs contain heterogeneous geometries and dense regions where 2D projected overlap does not reliably reflect actual spatial feasibility. As a result, the placement engine must reduce total wire-area while mitigating inconsistencies between 2D projection and true geometric feasibility. Consequently, the PCB placement problem addressed in this work is defined as follows:

- **2.5D-Aware Analytical Placement for Non-Planar Printed Circuit Board Designs.** Given a set of movable and fixed components, their geometric descriptions (e.g., STEP/OBJ models), a netlist, and spacing rules, the goal is to determine the 2D locations and orientations of all components on a multi-height PCB such that the total wire-area is minimized while reducing geometric infeasibility under the given spacing and design rules.

3 Proposed Algorithms

This section presents the proposed analytical placement framework for 2.5D non-planar PCB designs. Fig. 3 shows the overall flow of our framework. The framework first constructs mesh-based component models from mechanical geometry and projects them onto the 2D placement domain. During global placement, the density grid is reused as a spatial index to identify projected component interactions, and collision checks are applied only to candidate pairs that share density bins. The resulting 2.5D collision-aware density model distinguishes true physical collisions from legal vertical nesting. Finally, a 2.5D-aware ADMM legalization stage further refines the placement while preserving the same collision interpretation.

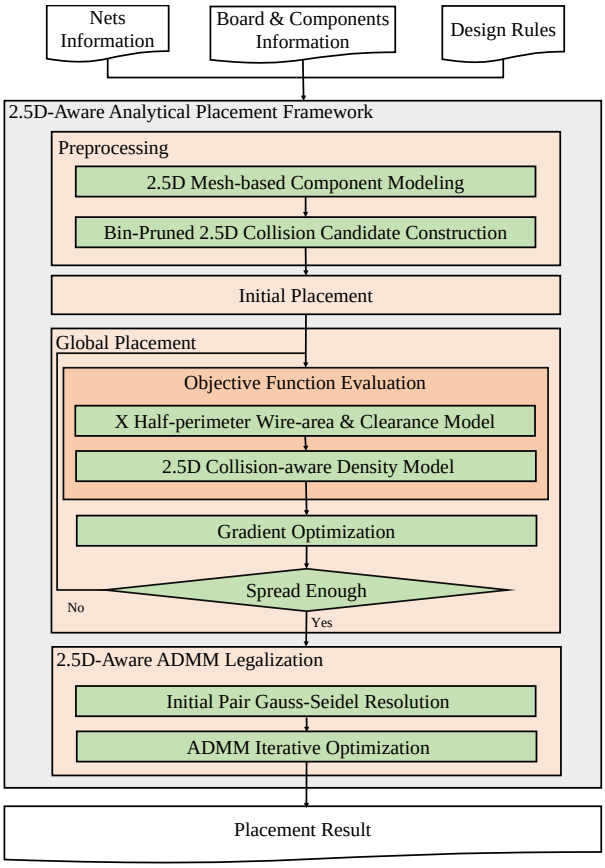


Figure 3: Overall flow of the proposed 2.5D-aware analytical placement framework.

3.1 Preprocessing

3.1.1 2.5D Mesh-based Component Modeling. Modern PCB modules increasingly employ multi-height components, irregular housings, and non-planar mechanical structures. To accurately reflect these characteristics during placement, each component is represented using a 3D surface mesh extracted from design files such as Standard for the Exchange of Product Data (STEP) or Wavefront OBJ (OBJ). As illustrated in Fig. 4, the raw mechanical model is converted into a set of triangular faces that describe the external envelope of the component. Only the outermost triangles are retained, allowing us to model the full 3D shape without the overhead of interior geometry.

For analytical placement, these mesh models must be related to the 2D optimization domain. Each mesh is first rigidly transformed according to the component's planar location and orientation, and then projected onto the xy-plane to obtain an effective 2D footprint. This projected polygon contributes area to the discretized density grid and identifies the regions where potential projected overlaps may occur. Using footprints derived from 3D meshes—rather than idealized library polygons—preserves the geometric irregularities of real components and enables more accurate interaction modeling during optimization.

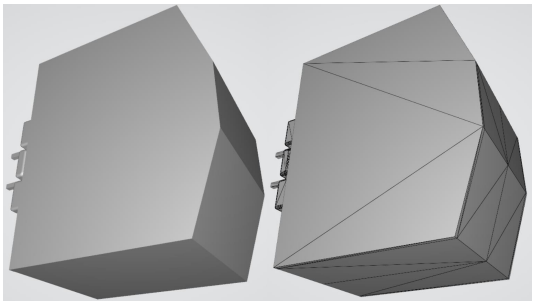


Figure 4: Illustration of mesh-based component modeling. The original mechanical model (left) is converted into a triangular surface mesh (right), from which only the outermost faces are retained to represent the component's 3D envelope for placement.

3.1.2 Bin-Pruned 2.5D Collision Candidate Construction. Directly checking all component pairs with mesh-level 3D collision tests is computationally prohibitive inside an analytical placement loop. To reduce this cost, we reuse the density grid as a spatial index for candidate-pair construction. After each component is transformed according to its current 2D location and orientation, its projected footprint is rasterized onto the density bins. For each bin, we collect the set of components contributing projected area to that bin. Only component pairs that appear in at least one common bin are inserted into the candidate set for 3D collision evaluation.

This bin-pruned construction avoids unnecessary pairwise tests between spatially distant components. It also aligns naturally with the density model: only projected interactions that may affect bin density are considered for 2.5D-aware exemption or penalty computation. As a result, the collision pipeline consists of a *broad phase*—a coarse spatial filter based on shared density bins that only identifies which pairs might interact—and a *narrow phase*—the detailed geometric test that actually determines whether a candidate pair collides in 3D.

For a candidate pair (i, j), we first apply inexpensive geometric filters, including bounding-box and vertical z-slab separation tests: if the two components' vertical (height) extents do not overlap at all, the pair is immediately treated as non-colliding in 3D and its projected overlap can be exempted without invoking the narrow phase. Otherwise, the pair is passed to the narrow-phase collision test. In our implementation, component meshes are preprocessed into convex parts using the existing V-HACD algorithm [24], and collision between convex pieces is evaluated using the existing Gilbert–Johnson–Keerthi (GJK) algorithm [25]; both are established techniques that we adopt rather than contribute, applying them here specifically to PCB component meshes under the density model's bin-pruned candidate set. This provides a practical balance between geometric accuracy and runtime efficiency for repeated use inside global placement.

Algorithm 1 summarizes the overall placement pipeline described in this section, from mesh-based candidate construction through the 2.5D-aware density update used during global placement.

Complexity. Let $n = |C|$ be the number of components, let k denote the average number of density bins touched by a component footprint, and let m_b denote the number of components assigned to bin b. Rasterizing all projected footprints costs $O(nk)$. Candidate construction enumerates component pairs within each occupied bin and therefore costs

$$O\left(\sum_{b \in B} m_b^2\right), \tag{6}$$

Algorithm 1 2.5D-Aware Global Placement

Require: Components C, meshes $\{M_i\}_{i\in C}$, bin grid B, positions $\mathbf{x}^{(0)}$
Ensure: Legalized positions $\mathbf{x}^*$
for iteration $t = 0, 1, \ldots$ until converged **do**
  Transform and rasterize each component's projected footprint onto B
  Build candidate pairs $\mathcal{P}_t$: pairs sharing at least one bin
  for each $(i, j) \in \mathcal{P}_t$ **do**
    if bounding-box or z-slab test separates i, j **then**
      Mark (i, j) as non-colliding; exempt overlap in ρ_b
    else
      Use precomputed V-HACD convex parts of M_i, M_j; run GJK on candidate convex-part pairs
      if no true intersection **then**
        Exempt overlap in ρ_b per Eq. (9)
  Solve Poisson equation for ϕ from corrected ρ; compute $\mathbf{E} = -\nabla\phi$
  Compute $\nabla W, \nabla S, \nabla D$; update $\mathbf{x}^{(t+1)}$
Run 2.5D-aware ADMM legalization on $\mathbf{x}^{(t)}$ to obtain $\mathbf{x}^*$
return $\mathbf{x}^*$

with duplicate pairs removed when the same pair shares multiple bins. In the pathological case where all components occupy the same bin, this term is $O(n^2)$; under the locally sparse occupancy observed in practical placements, however, m_b remains small and the candidate set is substantially smaller than exhaustive all-pairs testing. Each candidate pair is first screened by an $O(1)$ bounding-box/z-slab test. Only pairs that pass this filter invoke the narrow-phase V-HACD/GJK test, whose cost depends on the number of convex parts in the corresponding meshes. Convex decompositions are precomputed, and collision results are reused when the relative placement remains within the caching tolerance described below. Thus, the broad phase does not change the worst-case quadratic bound, but it reduces the number of expensive mesh-level tests to spatially relevant component pairs in typical PCB placements.

3.2 Initial Placement

Before global optimization, the solver initializes movable components inside the legal placement region. The initialization follows a lightweight strategy that distributes elements near the center of the board. Only coarse placement is required at this stage, as subsequent gradient optimization will refine the layout extensively. This step ensures that all components begin within valid boundaries and that no large-scale infeasibility is introduced before analytical spreading begins.

3.3 Global Placement

The global placement phase solves the continuous optimization problem of Eq. (1) over the 2D locations $\mathbf{x} = \{(x_i, y_i) \mid i \in C_m\}$ of all movable components, where $W(\mathbf{x})$ is the total wire-area, $S(\mathbf{x})$ is the pad-to-pad clearance penalty, and $D(\mathbf{x})$ is the 2.5D-aware density penalty with exemption terms. The weights λ_1 and λ_2 adjust the contributions of the clearance and density terms in the objective.

3.3.1 X Half-perimeter Wire-area & Clearance Model. PCB routing commonly follows 135° patterns, making the XHPWA a more faithful first-order estimate of routing span than conventional Manhattan HPWL. Following the clearance-aware formulation introduced in Chen *et al.* [7], our method computes an XHPWA-based routing footprint for each net $e \in N$ and modulates it with a clearance-dependent adjustment factor α. The factor α increases when the pads of e lie in locally crowded regions or when their separations approach spacing constraints, thereby reflecting the increased routing demand.

The wire-area contribution of each net becomes

$$W_e(\mathbf{x}) = \alpha \cdot w_e \cdot \mathrm{XHPWL}(e), \tag{7}$$

and the total wire-area is

$$W(\mathbf{x}) = \sum_{e\in N} W_e(\mathbf{x}). \tag{8}$$

A smooth pad-to-pad clearance penalty is additionally included to discourage placements that violate spacing rules. This unified model, inspired by [7], promotes early spreading around routing-critical nets and works jointly with the density term to guide the global optimization process.

3.3.2 2.5D Collision-Aware Density Model. The density formulation follows the classical electrostatic model, where the placement region is discretized into uniform bins and each component contributes area to the grid. The electrostatic potential derived from this density drives smooth spreading forces during optimization. Our formulation extends this model to properly account for 3D geometry and volumetric feasibility.

Projected footprint from 3D geometry. Instead of relying on simplified library polygons, each component contributes density according to the projected area of its transformed 3D mesh. The mesh is rigidly rotated and translated according to its placement parameters, and its triangular surface is projected onto the xy-plane to obtain a 2D footprint. This captures fine mechanical details and irregular outlines that become important under tight packing or non-orthogonal orientations, ensuring that the density field reflects the true physical envelopes of the components.

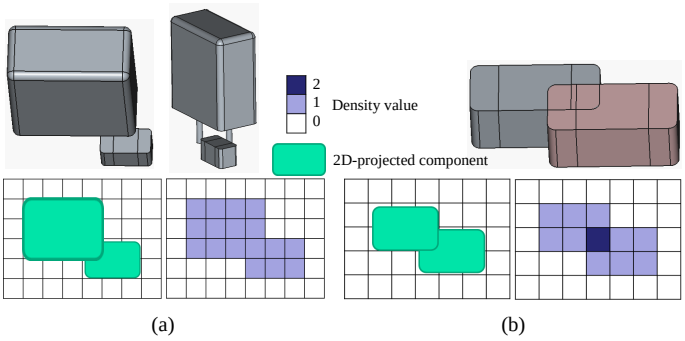


Figure 5: Effect of the 2.5D-aware exemption mechanism. (a) Two components overlap in 2D but remain separated in 3D, so only one density value is counted. (b) A true 3D collision causes the projected density to accumulate multiple values.

2.5D-aware exemption of benign overlaps. Because 3D components may legally interlock or be vertically separated, their 2D projections may overlap even when no physical collision exists. Treating all projected overlaps as density overflow would introduce false repulsive forces and unnecessarily restrict the solution space, so we incorporate a per-bin 2.5D-aware exemption mechanism instead.

For each bin b, every candidate pair (i, j) that contributes projected area to b is evaluated by the 3D collision pipeline. If the pair is verified to be non-colliding in 3D, the corresponding projected overlap is exempted from the bin density:

$$\rho_b \leftarrow \max\big(0,\ \rho_b - O_{ij}(b)\big), \tag{9}$$

where $O_{ij}(b)$ denotes the normalized exempted overlap of pair (i, j) in bin b. This update is applied pairwise. For bins containing three or more mutually overlapping components, pairwise overlap regions can interact and the corrected density should therefore be interpreted as an optimization surrogate rather than an exact union-volume measure. Section 4.4 discusses this limitation and the final legalization-stage re-evaluation.

If a true 3D collision is detected, the projected overlap remains in the density map and contributes to the electrostatic overflow. Therefore, legal vertical nesting does not generate artificial spreading force, whereas true volumetric interference is still penalized. This corrected density map is then used to solve the Poisson equation and compute the density gradient in the same manner as the original electrostatic placement engine.

Implementation-level acceleration. To reduce the overhead of repeated 3D collision evaluation, our implementation uses several acceleration mechanisms, each governed by a fixed, justified parameter held constant across all evaluated designs. First, vertical z-slab filtering skips the narrow phase for pairs whose height ranges are separated by more than 0.3 mm, matching the general-net clearance rule cited in Section 4.3. Second, collision results are cached per pair and reused within a 0.5 mm position/orientation tolerance, a conservative bound on typical per-iteration movement. Third, full rebuilds occur only every 5 steps—consistent with that same 0.5 mm tolerance—with incremental updates in between. Finally, 2.5D-aware reasoning activates only once density overflow drops below 0.3, the point at which coarse spreading has typically already occurred.

Figure 5 illustrates this effect. In the left example, two components overlap in 2D but remain separated in 3D, so the density map counts only a single occupancy layer. In the right example, where the meshes physically intersect, density accumulates multiple values. This ensures that the model penalizes genuine clashes while avoiding exaggerated congestion in multi-height assemblies.

3.3.3 Gradient Optimization. The placement is refined through an iterative gradient-based optimization procedure that continuously adjusts the 2D locations of all movable components. At each iteration, the analytical derivatives of the wire-area, clearance, and 2.5D-aware density terms are evaluated to produce a composite force that determines the direction and magnitude of movement for every component. The wire-area gradient encourages shorter routes, the clearance gradient pushes components away from tight pad-to-pad interactions, and the density gradient provides smooth electrostatic repulsion that reduces projected overflow.

A key feature of this optimization process is that the gradients naturally reflect the outcome of the 2.5D-aware density exemption: bins whose projected overlap is exempted generate no spreading force, while bins corresponding to true 3D interference contribute stronger repulsion. This allows the optimization to prioritize resolving genuine volumetric clashes rather than artificial 2D congestion.

After each update, component positions are recomputed, bins are refilled, and the entire objective is evaluated again to obtain the next gradient field. Step sizes are adaptively controlled to ensure stable descent and to avoid oscillation in densely packed regions. The optimization terminates once the density map becomes sufficiently uniform and no bins exhibit excessive overflow.

3.4 2.5D-Aware ADMM Legalization

After global placement, we apply a legalization stage based on the alternating direction method of multipliers (ADMM) [26], an existing consensus-optimization technique that we adopt rather than contribute, using its standard scaled-dual-variable form. Each iteration alternates an x-update, which solves a quadratic program moving every component toward its global-placement position and the current consensus target (minimizing displacement only), a z-update, which separately resolves pairwise geometric overlap via minimum-translation-vector pushes, and a u-update, which accumulates the residual between the two. The penalty weight ρ coupling the x- and z-updates is a fixed constant for the entire run rather than an escalating schedule, following standard scaled-dual-variable ADMM practice; the loop terminates when the primal residual falls below a threshold, a fixed iteration budget is reached, or an early-stopping rule triggers on repeated lack of improvement in collision count. The legalizer thereby minimizes displacement from the global-placement solution while iteratively resolving component overlaps. To ensure consistency with the 2.5D-aware global placement model, the legalization stage uses the same 3D collision interpretation: projected overlaps that are volumetrically separable are exempted, while true 3D collisions are treated as violations.

Formally, each iteration performs the scaled-dual-variable ADMM updates

$$\mathbf{x}^{(k+1)} = \arg\min_{\mathbf{x}} \left\{ \tfrac{1}{2}\|\mathbf{x} - \mathbf{x}^{\mathrm{gp}}\|_2^2 + \tfrac{\rho}{2}\|\mathbf{x} - \mathbf{z}^{(k)} + \mathbf{u}^{(k)}\|_2^2 \right\}, \tag{10}$$

$$\mathbf{z}^{(k+1)} = \mathcal{R}_{\mathrm{ovlp}}\left(\mathbf{x}^{(k+1)} + \mathbf{u}^{(k)}\right), \tag{11}$$

$$\mathbf{u}^{(k+1)} = \mathbf{u}^{(k)} + \mathbf{x}^{(k+1)} - \mathbf{z}^{(k+1)}, \tag{12}$$

where $\mathbf{x}^{\mathrm{gp}}$ denotes the fixed global-placement solution used as the consensus target, ρ is the fixed penalty weight introduced above, and $\mathcal{R}_{\mathrm{ovlp}}(\cdot)$ denotes the pairwise minimum-translation-vector overlap-resolution operator described next, applied independently to each component pair still in true 3D collision. Because both terms in Eq. (10) are quadratic in $\mathbf{x}$, the x-update reduces to a per-component weighted average of the global-placement position and the current consensus target, with no overlap term entering the objective directly; all overlap resolution instead happens through the overlap-resolution operator $\mathcal{R}_{\mathrm{ovlp}}$ in Eq. (11). Eq. (12) then accumulates the running discrepancy between the displacement-minimizing x-update and the overlap-resolving z-update, which is what allows the two to be reconciled gradually across iterations rather than in a single step.

The z-update's minimum-translation-vector computation is a 2D geometric test applied to the components' projected footprints, distinct from the mesh-level V-HACD/GJK pipeline used during global placement (Section 3). For each pair still in true 3D collision, the resolver enumerates candidate separating axes derived from the two projected polygons' edges and, among the axes on which the projections still overlap, selects the axis with the smallest overlap depth; the corresponding direction and magnitude define the translation applied to move the pair out of 2D overlap. This mechanism itself is reused verbatim between the 2D baseline and the 2.5D-aware configuration described below; only the decision of *whether* to invoke it on a given pair differs.

Before ADMM iterations, we apply an initial pairwise Gauss–Seidel resolution step to reduce severe local collisions and provide a more stable starting point for the subsequent ADMM optimization. This step identifies all pairs with initial geometric overlap and, over 30 alternating-direction sweeps, resolves each overlapping pair by splitting the corrective displacement evenly (50/50) between the two components, so that neither component is pushed disproportionately far from its global-placement position before ADMM begins.

The 2D baseline and our method share the identical ADMM formulation, sweep counts, and all other parameters, differing only in a single configuration flag: disabled, every overlap check falls back to an unconditional true-collision result, reproducing the 2D baseline exactly; enabled, the legalizer's collision resolver is given a handle to the 3D component manager and preserves valid vertical nesting while resolving true mesh-level intersections. This ensures that the comparison in Table 3 isolates

Table 2: Statistics of PCB benchmarks.

Design	$\lvert C\rvert$	#Fixed	#Pads	$\lvert E\rvert$	#Layers
3D-PCB1	34	10	134	68	1
3D-PCB2	56	10	178	105	1
3D-PCB3	56	11	178	106	2
3D-PCB4	34	10	134	68	2
3D-PCB5	75	12	217	141	2
3D-PCB6	91	17	251	173	2
3D-PCB7	91	13	251	174	2

Table 3: Comparison between Chen et al., the ablated (w/o 2.5D-aware legalization) variant, and our full method.

Design	Chen et al. [7]					w/o 2.5D-Aware Legalization					Full Method (Ours)				
	XHPWA	#2D Ov.	#3D Col.	#Exempt.	RT (s)	XHPWA	#2D Ov.	#3D Col.	#Exempt.	RT (s)	XHPWA	#2D Ov.	#3D Col.	#Exempt.	RT (s)
3D-PCB1	438.43	6	6	0	21.13	383.88	5	5	0	62.72	399.38	14	3	11	71.45
3D-PCB2	480.59	11	11	0	28.10	446.86	12	12	0	94.04	426.56	41	1	40	118.83
3D-PCB3	602.70	25	25	0	26.27	563.94	12	12	0	42.88	559.94	55	0	55	70.96
3D-PCB4	490.67	4	4	0	21.58	498.15	14	14	0	82.40	516.99	18	4	14	71.95
3D-PCB5	645.71	38	38	0	27.73	675.41	46	46	0	52.89	646.59	69	9	60	89.52
3D-PCB6	772.77	50	50	0	35.13	807.68	67	67	0	93.58	748.09	89	4	85	143.00
3D-PCB7	498.57	39	39	0	27.27	549.42	29	29	0	56.42	546.89	72	1	71	108.98
Avg.	561.35	24.71	24.71	0	26.74	560.76	26.43	26.43	0	69.28	549.20	51.14	3.14	48.00	96.38
Comp. (vs. Chen)	1.00	1.00	1.00	–	1.00	1.00	1.07	1.07	–	2.59	**0.98**	2.07	**0.13**	–	3.61
Comp. (vs. w/o Leg.)	–	–	–	–	–	1.00	1.00	1.00	–	1.00	**0.98**	1.94	**0.12**	–	1.39

the effect of 2.5D collision awareness rather than confounding it with other implementation differences.

During ADMM iterations, some initially exempted pairs may become true collisions as components move independently. To improve robustness, the legalizer tracks the true-collision count across iterations and resets the early-stopping patience counter whenever it improves or newly regresses, up to a bounded number of regression resets (three in our implementation); beyond this bound, further regressions no longer reset patience, giving the legalizer additional iterations to resolve newly formed conflicts while still guaranteeing termination.

4 Experimental Results

We evaluated our method on seven anonymized, industry-grade PCB designs provided by an industrial partner. These designs contain heterogeneous components, irregular mechanical housings, and both single-sided and double-sided board configurations. We refer to them as 3D-PCB1–3D-PCB7. The design statistics are summarized in Table 2.

4.1 Comparison of 2D vs. Collision-Aware 2.5D

Table 3 reports the legalized placement results of the 2D baseline and the proposed full method (the ablated variant without 2.5D-aware legalization is discussed separately in Section 4.2). Since existing PCB analytical placers do not explicitly model 2.5D physical collisions, the 2D baseline follows the analytical density model of Chen *et al.* [7], the closest conventional 2D baseline, using the same ADMM legalization framework with 2.5D collision awareness disabled, so every projected 2D overlap is treated as a physical conflict. In contrast, our full method enables both the 2.5D collision-aware density model during global placement and 2.5D-aware collision checking during legalization, allowing the placer to distinguish valid vertical nesting from true physical collisions; the comparison therefore focuses primarily on physical feasibility rather than requiring the proposed method to improve every conventional 2D placement metric simultaneously.

Compared with Chen *et al.*, our method reduces the average number of true 3D physical collisions from 24.71 to 3.14, corresponding to 0.13× of the 2D baseline, or an 87% reduction. This is the primary physical-feasibility metric of our evaluation, since a projected 2D overlap does not necessarily correspond to a physical collision in a 2.5D PCB.

The proposed method also identifies an average of 48.00 projected overlaps per design that are physically separable and therefore exempted from collision penalties. This substantially larger number of projected overlaps (51.14 versus 24.71 for the conventional 2D baseline) should therefore not be interpreted as a degradation in physical legality; rather, it reflects the ability of the proposed framework to distinguish valid vertical nesting from true volumetric interference.

XHPWA is additionally reported to characterize the resulting placement and connection-length quality, but it is not used as the primary measure of physical quality. In particular, XHPWA alone cannot distinguish a physically legal vertically nested configuration from a layout in which components with identical or similar projected footprints are placed on top of one another and therefore exhibit small projected connection lengths despite physical collisions. The average XHPWA decreases from 561.35 to 549.20 (0.98×); per design, XHPWA improves for four of the seven designs (3D-PCB1, 3D-PCB2, 3D-PCB3, 3D-PCB6) and increases for the remaining three (3D-PCB4, 3D-PCB5, 3D-PCB7), as shown in Table 3, since enabling legal vertical nesting changes which placement solution the optimizer converges to and is not guaranteed to uniformly shorten 2D connection lengths on every design even as it improves physical feasibility. This aggregate XHPWA comparison should therefore be interpreted only as a secondary placement-quality indicator rather than a direct measure of 2.5D physical feasibility.

At the local level, Figure 1 provides a complementary example of the benefit enabled by 2.5D-aware placement. By allowing a physically valid vertical nesting relationship, the combined length of the illustrated red connections decreases from 12.97 mm to 6.51 mm, a 49.81% reduction. This reduction is a local consequence of exploiting legal nesting and should not be interpreted as the overall XHPWA improvement reported in Table 3. The difference between the two values arises because Figure 1 measures a specific set of connections in one placement example, whereas XHPWA aggregates connection-related placement quality over the complete design.

The average runtime of our full method is 96.38 s, compared with 26.74 s for the 2D baseline and 69.28 s for the ablated variant (discussed in Section 4.2), corresponding to 3.61× and 1.39× their runtimes, respectively. This increase mainly comes from V-HACD/GJK-based collision evaluation and additional 2.5D-aware legalization iterations. The runtime overhead represents the computational cost of explicitly evaluating 3D geometric feasibility and is therefore reported together with the physical-quality improvements rather than being treated as a placement-quality metric. The overhead scales primarily with the number of candidate pairs generated by the bin-pruned broad phase rather than with the total component count directly, since only spatially co-located pairs invoke the narrow-phase V-HACD/GJK test. As a result, congested designs with many overlapping projected footprints incur a larger share of the runtime increase than sparser designs with a comparable number of components. Figure 6 illustrates the resulting placement outcomes under the 2D and 2.5D collision-aware modes.

4.2 Ablation Study on 2.5D-Aware Legalization

The *w/o 2.5D-Aware Legalization* rows in Table 3 isolate the contribution of 2.5D-aware legalization by holding the proposed 2.5D-aware global placement density model fixed and disabling 2.5D-aware collision handling only during legalization. With this stage disabled, the average true collision count increases from 3.14 to 26.43, corresponding to an 8.4× regression, confirming that legalization is essential for resolving residual volumetric conflicts while retaining the 2.5D-aware placement decisions generated during global placement. This is consistent with the distinct roles of the two stages: global placement primarily provides coarse, gradient-driven spreading guidance, whereas legalization performs the fine-grained, pairwise adjustments needed to resolve residual

Table 4: Accuracy analysis: mesh-based vs. STEP-based distances.

Metric	Absolute error (mm)	Relative error (%)
Mean	0.016	1.02
Median	0.0004	0.002
95th percentile	0.041	0.25
Maximum	1.007	95.88

Table 5: Runtime comparison for distance and collision queries.

Modeling	Avg time (ms)	Median time (ms)
STEP	116.3	68.5
Mesh	57.6	0.75

conflicts that remain after global placement converges—conflicts that are left largely unresolved without 2.5D-aware legalization, even though the density model itself already distinguishes benign overlaps from true collisions.

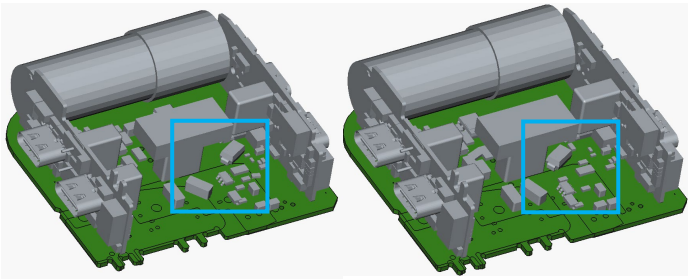


Figure 6: Illustration from the 3D-PCB4 benchmark. The component highlighted in the blue box in (a) is placed farther away under the 2D model because its projection falls within the neighboring T-shaped module. In (b), the 2.5D collision-aware model reveals usable vertical space in the same region, allowing the component to nest beneath the T-shaped module and move closer to its connection endpoints.

4.3 Accuracy and Runtime of Mesh-Based 3D Collision Evaluation

We evaluated both the geometric fidelity and computational efficiency of using triangular meshes in place of native STEP models for collision reasoning, using 90 PCB components across 100 randomly sampled component pairs and measuring the minimum distance between each pair under both representations. As summarized in Table 4, the mean absolute error is only 0.016 mm and the 95th-percentile error is 0.041 mm, both negligible compared to typical PCB clearance rules (0.20–0.30 mm for general nets, 3.3–4.2 mm for high-voltage nets). The large maximum error arises only in rare near-contact cases, which are handled conservatively by treating coplanar contact as collision.

Table 5 shows that mesh-based distance queries are also significantly faster than STEP-based ones: the average query time drops from 116.3 ms to 57.6 ms, and the median from 68.5 ms to only 0.75 ms, further amplified within our placement flow by bin-level pruning that limits 3D checks to spatially relevant pairs. Mesh-based collision evaluation therefore provides both sufficient geometric accuracy and practical runtime for repeated use inside the density loop.

4.4 Multi-Component Overlap and Nesting

The proposed 2.5D collision-aware density model does not assume that a density bin contains at most two components: every candidate pair contributing projected area to a given bin is evaluated independently against its own 3D geometry, and only that pair's overlap is exempted from the bin density if the pair is verified non-colliding (Eq. (9)).

This pairwise, additive exemption has a genuine limitation for dense bins containing three or more mutually overlapping components: if a single overhanging component can physically accommodate at most one of two smaller components underneath it at a time, both pairwise checks may independently report no collision, so the density model would exempt both overlaps and locally understate the true bin density even though only one can actually fit. This risk is confined to the smooth density signal that guides global placement and does not by itself certify the final layout—the 2.5D-aware legalization stage (Section 3) re-evaluates every candidate pair's exact 3D configuration independently of the density model's history and treats any pair still in true collision as a violation to be resolved. Because any point in a three-or-more-way intersection must lie in every pairwise intersection among them, this exhaustive pairwise re-check cannot miss a genuine multi-way collision at final positions, so a jointly infeasible arrangement the density model under-penalized during optimization is still caught and reported as a violation, ensuring that the residual collisions in Table 3 accurately reflect the true physical state of the final layout. Extending the exemption to reason jointly over three or more components is left to future work.

5 Conclusions

We have presented an analytical placement framework for 2.5D non-planar PCB designs. The proposed framework integrates mesh-level 3D geometric reasoning into an electrostatic density model by reusing the density grid for bin-level candidate pruning and applying V-HACD/GJK-based narrow-phase collision detection only to spatially relevant component pairs. For component pairs that overlap in 2D projection but are physically separable in 3D, our method applies a conservative density exemption, allowing the analytical placer to exploit legal vertical nesting without modifying the core Poisson-based optimization backbone. We further extend the legalization stage with 2.5D-aware collision handling to consistently preserve valid nesting while resolving true volumetric collisions. The framework also supports configurations involving multiple overlapping components by evaluating spatially relevant 3D interactions rather than assuming that each projected overlap corresponds to a physical collision.

Experimental results on seven industry-grade PCB benchmarks show that our method reduces true physical collisions to 13% of the 2D baseline and identifies substantial valid vertical nesting opportunities. The resulting increase in projected 2D overlaps is therefore not directly indicative of degraded physical quality, since many of these overlaps are physically separable in 3D. XHPWA is consequently treated as a secondary placement-quality metric rather than a standalone measure of physical feasibility.

The results demonstrate that explicit 2.5D geometric reasoning can substantially improve physical feasibility and enable tighter component arrangements by exploiting legal vertical nesting, while retaining the analytical placement optimization backbone, with additional placement runtime that remains practical on the evaluated benchmarks. Future work includes extending the per-bin exemption to reason jointly over three or more mutually overlapping components rather than pairwise, and evaluating the framework on larger, more diverse multi-height PCB assemblies.

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